kinetic_ORDERING

KINETIC motion controller + vibration puck: ordering guide (what to buy, where, in what order)

> Companion to `MANUFACTURING_MANUAL.md`. Main caveat for this project: never run on a motor; AI on synthetic data. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: `manuals/ORDERING_GUIDE.md`.

1. Order of work

1. Read the manual and the caveat above.
2. Order the circuit boards and SMD assembly (section 2).
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

2. Order the circuit boards (JLCPCB, PCB + assembly)

Files (all in this repository):

- Gerbers: `utility10/pcb/release/kinetic_ctl/1_fabrication/kinetic_ctl_gerbers.zip`
- Assembly BOM: `utility10/pcb/release/kinetic_ctl/2_assembly/kinetic_ctl_JLC_BOM.csv`
- Pick-and-place (CPL): `utility10/pcb/release/kinetic_ctl/2_assembly/kinetic_ctl_JLC_CPL.csv`
- Full BOM with LCSC numbers and prices: `costing/kinetic_ctl_priced_BOM.csv`

Steps:

1. Create an account at **jlcpcb.com** and click *Order now*. Upload the Gerber ZIP. The viewer must show the board outline, both copper layers, silkscreen and drill holes.
2. Settings: **2 layers, 1.6 mm, FR-4**, lead-free HASL (use **ENIG** if the board has an LGA/QFN fine-pitch part, e.g. ADXL345, ESP32 module pad, BQ25570), green mask, white silk, quantity 5 (the minimum), *Remove order number: specify a location* if you want to avoid the order-number mark. Heavier copper (2 oz) only where the board notes say so.
3. Switch on **PCB Assembly**: *Economic*, assembly side *Top*, and set tooling holes *Added by JLCPCB* unless the board already has them. Choose the number of boards to assemble (2 or 5).
4. Upload the BOM and CPL. In the parts step, every line must match (LCSC numbers are in the BOM). Where a part is shown as *out of stock* or *no match*, use the alternative suggested by JLCPCB only after checking footprint and value against the list below; do not accept a different footprint.
5. In the **component placement preview**, check pin 1 / polarity of every IC, diode, LED, electrolytic capacitor, USB-C and connector against `*_assembly_top.pdf` in the same folder. Wrong rotation is the most common assembly error: ask JLCPCB support to rotate before approving.
6. Pay, then approve the assembly engineering review messages (they usually ask about a part or rotation). Delivery by DHL/FedEx/UPS takes about 1 to 2 weeks to India.

Parts **not** assembled by JLCPCB (you buy and solder them yourself):

Refs	Value	Part	LCSC
J5	24 V in	KF128-5.08-2P-AA	C474952
J7	Host UART (Pi/PC)	JST B3B-PH-K-S(LF)(SN)	C131339
J3	Vibration sensor SPI	HCTL PZ254-2-03-Z-8.5	C2894964
J8	Endstop X	JST B3B-PH-K-S(LF)(SN)	C131339
J9	Endstop Y	JST B3B-PH-K-S(LF)(SN)	C131339
J10	Endstop Z	JST B3B-PH-K-S(LF)(SN)	C131339
TP1, TP2, TP3, TP4	TP		
J11	X driver logic	hanxia HX PM2.54-1x8P	C22438158
J12	X driver power/motor	hanxia HX PM2.54-1x8P	C22438158
J20	Motor X	JST B4B-PH-K-S(LF)(SN)	C131334
J13	Y driver logic	hanxia HX PM2.54-1x8P	C22438158
J14	Y driver power/motor	hanxia HX PM2.54-1x8P	C22438158
J21	Motor Y	JST B4B-PH-K-S(LF)(SN)	C131334
J15	Z driver logic	hanxia HX PM2.54-1x8P	C22438158
J16	Z driver power/motor	hanxia HX PM2.54-1x8P	C22438158
J22	Motor Z	JST B4B-PH-K-S(LF)(SN)	C131334

Order these with the same LCSC order (add them to the cart so one shipment arrives) or buy locally in India (section 5).

Parts cost and stock (live LCSC prices on the run date)

Total parts per board: **US8.49*for*1*board* * *, * * *US*7.72 each for 10 boards**, **US$6.21 each for 100 boards** (about INR 747 / 679 / 546 at 88 INR/USD). The per-line break-up is in `costing/kinetic_ctl_priced_BOM.csv`. This is parts only: PCB fabrication, assembly fees, shipping, customs and GST come on top (see `manuals/ORDERING_GUIDE.md`, section 6).

Stock warnings on the run date:

- TP (no lcsc price (hand-fit or not priced), stock)

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
J5	1	24 V in	KF128-5.08-2P-AA	C474952	0.1948	0.195
D3	1	SMBJ28A	Littelfuse SMBJ28A	C178227	0.1091	0.109
Q5	1	AO3401A	AO3401A	C15127	0.0969	0.097
D4	1	BZT52C12	DIODES BZT52C12-7-F	C124196	0.0775	0.077
R40,R42,R47,R51,R52,R53	6	10k	0603 10k 1%	C25804	0.0024	0.014
C30	1	100u/35V	Lelon VZH101M1VTR-0607	C176683	0.0881	0.088
C31	1	10u/50V	Samsung CL32A106KBJNNNE	C380537	0.2086	0.209
U2	1	AP63205WU-7	AP63205WU-7	C2071056	0.4665	0.467
R41,R46	2	100k	0603 100k 1%	C25803	0.0030	0.006
C32,C37,C40,C41,C42,C43	6	100n	Samsung CL10B104KB8NNNC 0603 100nF X7R 50V	C1591	0.0075	0.045
L1	1	4.7u	Changjiang FNR5040S4R7MT	C167963	0.0933	0.093

Refs	Qty	Value	Part	LCSC	Unit USD (1 board)	Line USD
C33,C34	2	22u/10V	Samsung CL21A226MPQNNNE	C29277	0.0377	0.075
D5	1	SS14	SS14	C84343	0.0252	0.025
U3	1	AP2112K-3.3	AP2112K-3.3TRG1	C51118	0.1814	0.181
C35,C39	2	1u	0603 1uF X7R	C15849	0.0144	0.029
C36,C38	2	10u	0805 10uF X5R	C15850	0.0441	0.088
U1	1	ESP32-S3-WROOM-1-N8	ESP32-S3-WROOM-1-N8	C2913198	4.9100	4.910
SW2	1	Reset (EN)	Omron B3U-1000P	C231329	0.1894	0.189
SW3	1	Boot (IO0)	Omron B3U-1000P	C231329	0.1894	0.189
J1	1	USB-C	TYPE-C-31-M-12	C165948	0.1890	0.189
R43,R44	2	5.1k	0603 5.1k 1%	C23186	0.0016	0.003
D6	1	Green	Hubei KENTO KT-0603YG yellow-green 570 nm, Vf about 2 V (lights well from 3.3 V with 1 k; the Everlight 19-217 green was out of stock)	C2289	0.0105	0.011
R45,R48,R49,R50	4	1k	0603 1k 1%	C21190	0.0017	0.007
J7	1	Host UART (Pi/PC)	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J3	1	Vibration sensor SPI	HCTL PZ254-2-03-Z-8.5	C2894964	0.0766	0.077
J8	1	Endstop X	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J9	1	Endstop Y	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
J10	1	Endstop Z	JST B3B-PH-K-S(LF)(SN)	C131339	0.0420	0.042
TP1,TP2,TP3,TP4	4	TP				
J11	1	X driver logic	hanxia HX PM2.54-1x8P	C22438158	0.1359	0.136
J12	1	X driver power/motor	hanxia HX PM2.54-1x8P	C22438158	0.1359	0.136
J20	1	Motor X	JST B4B-PH-K-S(LF)(SN)	C131334	0.0458	0.046
J13	1	Y driver logic	hanxia HX PM2.54-1x8P	C22438158	0.1359	0.136
J14	1	Y driver power/motor	hanxia HX PM2.54-1x8P	C22438158	0.1359	0.136
J21	1	Motor Y	JST B4B-PH-K-S(LF)(SN)	C131334	0.0458	0.046
J15	1	Z driver logic	hanxia HX PM2.54-1x8P	C22438158	0.1359	0.136
J16	1	Z driver power/motor	hanxia HX PM2.54-1x8P	C22438158	0.1359	0.136
J22	1	Motor Z	JST B4B-PH-K-S(LF)(SN)	C131334	0.0458	0.046

3. Bought-in and mechanical items

Bought-in

StepStick driver modules (1-3), NEMA17 motors, 12-28 V supply (24 V recommended) with current limit capability for bring-up, 1x8 female socket strips are on the BOM, endstop switches, 6-wire cable for the puck (under 30 cm),

M2.5 screws.

4. Confidence (read before you order)

- **Order confidence: High**: all parts live-checked, none flagged.
- **Works-as-intended confidence: Medium**: ESP32-S3 controller with three StepStick sockets and a 24 V to 5 V buck. Motion library (370 checks) and protocol are host-tested; the ESP32 firmware has never been compiled; drivers and motors are bought in. The vibration model is trained on synthetic data only.
- **First thing to do:** Power with a current-limited supply and no drivers; then one driver and one motor at low current.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts	**LCSC** (lcsc.com) with the same cart as JLCPCB
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.